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To cite this article: Cheng-Wei Ma, Zhi-Long Xiu & An-Ping Zeng (2015) Exploring signal transduction in heteromultimeric protein based on energy dissipation model, Journal of Biomolecular Structure and Dynamics, 33:1, 134-146, DOI: [10.1080/07391102.2013.855145](https://doi.org/10.1080/07391102.2013.855145)

To link to this article: <https://doi.org/10.1080/07391102.2013.855145>



Published online: 27 Nov 2013.



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Exploring signal transduction in heteromultimeric protein based on energy dissipation model

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Communicated by Ramaswamy H. Sarma

(Received 26 September 2013; accepted 10 October 2013)

Dynamic intersubunit interactions are key elements in the regulation of many biological systems. A better understanding of how subunits interact with each other and how their interactions are related to dynamic protein structure is a fundamental task in biology. In this paper, a heteromultimeric allosteric protein, *Corynebacterium glutamicum* aspartokinase, is used as a model system to explore the signal transduction involved in intersubunit interactions and allosteric communication with an emphasis on the intersubunit signaling process. For this purpose, energy dissipation simulation and network construction are conducted for each subunit and the whole protein. Comparison with experimental results shows that the new approach is able to predict all the mutation sites that have been experimentally proved to desensitize allosteric regulation of the enzyme. Additionally, analysis revealed that the function of the effector threonine is to facilitate the binding of the two subunits without contributing to the allosteric communication. During the allosteric regulation upon the binding of the effector lysine, signals can be transferred from the β -subunit to the catalytic site of the α -subunit through both a direct way of intersubunit signal transduction, and an indirect way: first, to the regulatory region of the α -subunit by intersubunit signal transduction and then to the catalytic region by intramolecular signal transduction. Therefore, the new approach is able to illustrate the diversity of the underlying mechanisms when the strength of feedback inhibition by the effector(s) is modulated, providing useful information that has potential applications in engineering heteromultimeric allosteric regulation.

Keywords: protein dynamics; energy dissipation model; intersubunit signal transduction; allosteric communication; aspartokinase

1. Introduction

Multisubunit proteins are molecules whose functions almost invariably depend on intersubunit interactions, and these interactions are affected in physiologically important ways by sometimes subtle, sometimes striking changes in protein conformation. The functions of multisubunit proteins involve interactions among a variety of different subunits. Sometimes, these interactions are fleeting, though they may be the basis of complex physiological processes such as oxygen transport, immune function, and muscle contraction. On the other hand, proteins are dynamic molecules (Benkovic & Hammes-Schiffer, 2003; Cecchini, Houdusse, & Karplus, 2008; Kolomeisky & Fisher, 2007; Olsson, Parson, & Warshel, 2006). The dynamic character can be the result of internal conformational dynamics in the constituent molecules, exchange processes involving alternative complexes, or simply the equilibrium between free and bound forms. Dynamic processes are potential candidates to be perturbed by small molecules as the energy difference between alternative states may be of the same order of magnitude as the binding energy of small

ligands. A better understanding of how subunits interact with each other and how their interactions are related to dynamic protein structure is a fundamental task in biology.

In our recent work (Ma, Xiu, & Zeng, 2011), a novel approach to reveal intramolecular signal transduction network was proposed. A new algorithm of network construction was developed, which is based on a new protein dynamics model of energy dissipation (Ma, Xiu, & Zeng, 2012). The well-studied allosteric enzyme, *Escherichia coli* aspartokinase III, was used as a model system to demonstrate the new method. A key feature of this approach is that the direction of interaction is specified after inferring protein residue–residue interaction (RRI) network involved in the process of signal transduction. This enables fundamental analysis of the regulation hierarchy and identification of regulation hubs of the signaling network. In this paper, a heterotetrameric allosteric protein, *Corynebacterium glutamicum* aspartokinase, is used as a model system to explore intersubunit interactions and the signal transduction network based on the energy dissipation model.

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In micro-organisms and plants, lysine, threonine, and methionine are produced from aspartate. The enzyme catalyzing the first step of this pathway is aspartokinase (AK; EC 2.7.2.4), which phosphorylates the β -carboxyl group of aspartate using ATP. As found in enzymes catalyzing the first reactions involved in several other amino acid biosynthetic pathways, AK is regulated through feedback inhibition by the end products. Most organisms have several AK isozymes that are regulated differently. For example, *E. coli* possesses three different AKs. Isoforms I and II have aspartokinase-homoserine dehydrogenase (AK-HSDH) I and II activities (Cohen, Stanier, & Le Bras, 1969). AK-HSDH I is both inhibited allosterically and repressed by threonine, whereas AK-HSDH II is repressed by methionine (Falcoz-Kelly et al., 1972; Falcoz-Kelly, van Rapenbusch, & Cohen, 1969; Veron, Falcoz-Kelly, & Cohen, 1972). AK III, in contrast, is monofunctional and is allosterically inhibited by lysine (Cassan, Parsot, Cohen, & Patte, 1986; Stadtman, Cohen, Lebras, & de Robichon-Szulmajster, 1961). *C. glutamicum* has only a single AK (CgAK), but is regulated by threonine and lysine in a concerted manner (Shiio & Miyajima, 1969). All AKs whose crystal structures have been revealed so far are homooligomeric AKs; however, CgAK has a unique heteromultimeric structure, which is composed of equimolar α - and β -subunits encoded by in-frame overlapping genes (Kalinowski, Bachmann, Thierbach, & Pühler, 1990; Kalinowski et al., 1991). In CgAK, the amino acid sequence of the β -subunit is the same as that of the C-terminal domain of α -subunit. The N-terminal portion of α -subunit functions as the catalytic domain, whereas the β -subunit and the C-terminal portion of α -subunit serve as the regulatory domain (Yoshida, Tomita, Kuzuyama, & Nishiyama, 2010) (Figure 1).

The RRIIs involved in the allostery of CgAK has been recently studied by a combined approach of evolutionary statistic analysis and molecular dynamics simulation (Chen, Rappert, Sun, & Zeng, 2011). Several

mutations have been successfully developed to alter the allostery of CgAK by the effectors. The application of the energy dissipation model to the CgAK revealed hereto unknown intersubunit dynamic features and signaling network in this heteromultimeric enzyme.

2. Materials and methods

The methods and procedures for the energy dissipation simulation and network construction were described in detail in our previous work (Ma et al., 2011, 2012). They are adapted in this work as briefly described below.

2.1. Structure preparation

The X-ray diffraction structures of CgAK (PDB code: 3AB2), which retrieved from Protein Data Bank (Berman et al., 2000), was neutralized by adding sodium and chlorine ions with an ionic concentration of $.5 \text{ mol L}^{-1}$ and solvated in a rectangular box of TIP3P (Jorgensen, 1982) water molecules with a minimum solute-wall distance of 10 Å. The solvated system was energy-minimized by 5000 steps employing the software of NAMD (Phillips et al., 2005) prior to the molecular dynamics simulations in order to relax the loops and side chains to make them suitable for performing the simulations.

2.2. Molecular dynamics simulations

Molecular dynamics simulations were performed with a periodic boundary condition in the NPT ensemble using Langevin dynamics at 310 K with the damping coefficient of 5.0 ps^{-1} and constant pressure of 1 atm. The non-bond pair list was updated every 10 steps and the Particle Mesh Ewald (PME) method (Ryckaert, Ciccotti, & Berendsen, 1977) was used to treat long-range electrostatic interactions. A residue-based cut-off of 12 Å was applied to the non-covalent interactions. No constraint was applied to the protein during the molecular

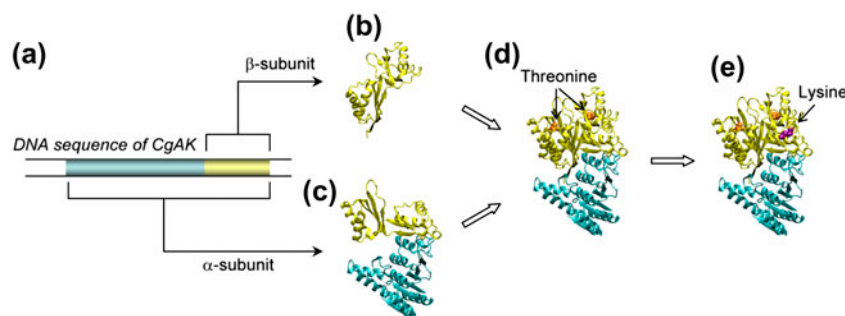


Figure 1. Structure of CgAK. (a) Diagram of the DNA sequence of CgAK. (b) β -subunit of CgAK. (c) α -subunit of CgAK. N-terminus of the α -subunit is colored in cyan and C-terminus of the α -subunit is colored in yellow. (d) CgAK with effector threonines. (e) CgAK with effector threonines and lysine. The regulatory domain of CgAK is composed of β -subunit and the C-terminus of α -subunit; the catalytic domain of CgAK is composed of the N-terminus of α -subunit.

dynamics simulations. A time step of 2 fs was used and the coordinates of the simulated complexes were saved every 1 ps. The simulations lasted 1 ns and were performed employing the software of NAMD with the CHARMM27 force field. To examine the convergence of the molecular dynamics simulations, energy, temperature, and pressure were monitored during simulations. The backbone root-mean-square deviations (RMSD) along the trajectory were also used to ensure the dynamical stability of the systems.

2.3. Energy dissipation simulations

In the energy dissipation simulations, the energy of key residues residing at the regulatory site was changed via increasing its velocity by three times. The energy dissipation simulations were conducted with a time step of 1 fs under the condition that the temperature and pressure of the system were able to change automatically during the simulation process (NVE ensembles). Other simulation parameters were the same as that used in the former step. The energy of each amino acid residue was captured during the entire dissipation process and the simulations lasted till all residues had responded to the energy perturbation. In the meanwhile, reference simulations were conducted to reduce the influence of the use of different structures from the former step. In reference simulations, the velocities of the residues were not changed to simulate the molecular dynamics process when no external energy was input to the protein.

2.4. Network construction

With a cut-off value for the distance of residue interactions, an undirected RRI network can be obtained from the crystal structure of a protein. In this work, a residue is considered to interact with those whose atoms are within a cut-off distance of 6 Å. Data of residue response time obtained from the energy dissipation simulations were then put into an undirected network, resulting in a directed RRI network (referred to as the Initial network). Next, the signal transduction network from the source residue to the target residue was achieved using the algorithm which consists of three major steps:

Step 1. Nodes whose response time is larger than t (the response time of the target residue) are deleted.

Step 2. Nodes whose in-degree is zero are deleted except the source residue. This step cycles until the number of nodes of the new generated network does not change any more. The value of node in-degree is re-calculated for each new network.

Step 3. Nodes whose out-degree is zero are deleted except the target residue. This step cycles until the number of nodes of the new generated network does not change any more. The value of node out-degree is recalculated for each new network.

Since the regulatory site and the catalytic site are always composed of several key residues that interact with ligand, the procedure was conducted for each pair of source–target residues to get the signal transduction networks from the regulatory site to the catalytic site. The obtained networks are referred to as the source–target pair networks.

2.5. Characterization of the energy dissipation process

To describe the characteristics of the energy dissipation process within proteins, the two-state Boltzmann equation is employed:

$$NR = \frac{NR_1 - NR_2}{1 + \exp[(\Delta t - t_0)/dt]} + NR_2 \quad (1)$$

$$NR' = (NR_2 - NR_1)/(4 \times dt) \quad (2)$$

where Δt is the response time, NR is the number of residues affected by external energy perturbation during the energy dissipation process, t_0 is the time point when half of the total number of residues have been affected, dt is a time constant, NR_1 is the Initial number of affected residue(s), NR_2 is the final number of affected residues, and NR' is the first derivative at $(t_0, (NR_1 + NR_2)/2)$, which reflects the change rate of the number of affected residues at this time point.

2.6. Definition of intermolecular propagation coefficient

To characterize the information propagation capability of a node residing at the interface of the signal transduction network, a new parameter of intermolecular propagation coefficient (IPC) is defined as follows:

$$IPC(i) = \frac{m_i \times n_i}{2 \times \sum_{j=1}^k m_j \times n_j} \quad (3)$$

where $IPC(i)$ is the intermolecular propagation coefficient of node i ; k is the total number of nodes residing in the intermolecular network; m_i and m_j are the in-degree of node i and j ; and n_i and n_j are the out-degree of node i and j . The denominator of the right side stands for all possible pathways between the molecules. The number 2 is added to remove the duplication of edge counting when calculating both in- and out-degree for all nodes. The numerator stands for the possible pathways conducted by node i .

3. Results

3.1. Energy dissipation processes within the α - and β -subunits from their threonine binding pockets

The energy dissipation curve describes the characteristic of the energy dissipation process by the number of residues that has responded to the energy perturbation at

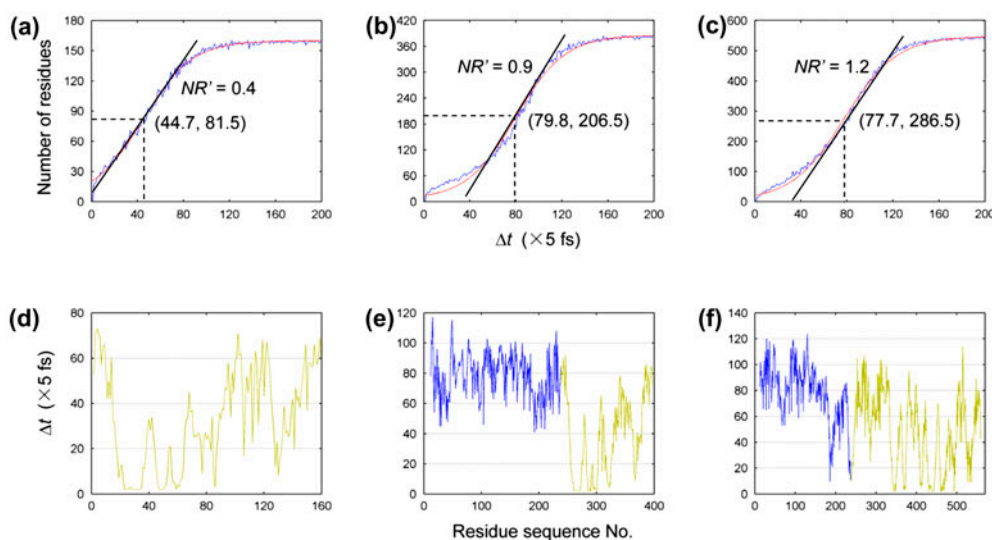


Figure 2. Energy dissipation curve and residue response time. Energy dissipation curve when residues residing at the threonine binding pocket of β -subunit (a), α -subunit (b) are perturbed and when residues residing at the lysine binding pocket of CgAK are perturbed (d). Energy dissipation curves are shown in blue lines; results of the two-state Boltzmann equation are shown in red lines. Residue response time when residues residing at the threonine binding pocket of β -subunit (d), α -subunit (e) are perturbed and when residues residing at the lysine binding pocket of CgAK are perturbed (f). Residues of the regulatory domain are shown in yellow lines; residues belonging to the catalytic domain are shown in blue lines.

Table 1. Parameters of the two-state Boltzmann equation for description of the energy dissipation process.

Proteins	NR_1	NR_2	t_0 (fs)	dt (fs)	NR' (residues fs^{-1})	Adj- R^2
β -subunit	3	160	223.7	106.3	.4	.995
α -subunit	3	408	399.2	106.6	.9	.994
CgAK	5	570	388.7	113.8	1.2	.997

different time points. Figure 2(a) and (b) shows the results when energy was put into the key residues of the threonine binding pocket (Asp274, Ala279, and Gln298 for the α -subunit and Asp25, Ala30, and Gln49 for the β -subunit). The energy dissipation curve usually exhibits a sigmoid pattern, which can be divided into three stages as given by the α -subunit (Figure 2(b)). During the first and third stage, the protein residues respond to the energy perturbation with a linear rate, while with an exponential rate in the second stage. However, the energy dissipation process of the β -subunit shows a non-typical sigmoid curve that lacks the first stage (Figure 2(a)). In our previous study (Ma et al., 2011), the two-state Boltzmann equation was proved to characterize the energy dissipation curve satisfactorily and thus it was used here (Table 1). The parameter t_0 in Equation (1), which is defined as the ‘half response time,’ is the time point when half of the residues have responded to the energy perturbation. For the β -subunit, half of the residues had been influenced at the time point of 223.7 fs, whereas the half response time of the α -subunit

is 399.2 fs because of its large number of residues. The first derivative (NR') of the energy dissipation curve at the time point of half response time, termed as ‘dissipation rate constant’, describes the exponential rate of the energy dissipation process. The dissipation rate constant of the α -subunit (.9 residues fs^{-1}) is larger than that of the β -subunit (.4 residues fs^{-1}) by over twofold, which means the dissipation process of the energy perturbation is maximally twofold faster within the α -subunit than that within the β -subunit.

The response time of each amino acid residue was calculated to get information at the level of residues (Figure 2(d) and (e)). In Figure 2(e), the curve is colored according to different structural regions of the α -subunit. Residues residing at the binding surface of the two subunits were recognized and their response time was given in Tables 2 and 3. It can be seen that the response time of residues residing at the binding surface of the β -subunit ranges from 10 fs for Val47 and Asn50 to 265 fs for Met1. In the case of the α -subunit, it ranges from 10 fs for Val296, Asn299, and Thr311 to 385 fs for Gly150.

Table 2. Response time of the target residue, the number and distribution of nodes in the different source-target pair signaling networks of the β -subunit.

Source-target pair networks ^a	Residue response time (fs)	Number of nodes	Distribution of the nodes in the source-target pair networks										
			1 ^b	2	3	4	5	6	7	8	9	10	11
Met1	265	10	1	1	0	2	0	2	0	2	1	1	0
Arg35	15	6	0	0	2	0	1	0	3	0	0	0	0
Asn43	130	28	0	3	2	6	1	8	4	2	1	1	0
Val47	10	2	0	0	0	0	0	0	0	0	1	1	0
Asn50	10	2	0	0	0	0	0	0	1	0	0	1	0
Thr62	30	16	0	0	0	0	0	8	4	2	1	1	0
Pro109	225	47	1	2	14	11	1	10	4	2	1	1	0
Glu117	245	48	2	2	14	11	1	10	4	2	1	1	0
Asn125	205	61	11	4	16	11	1	10	4	2	1	1	0
Ile129	115	21	0	0	0	3	0	10	4	2	1	1	0
Thr131	75	6	0	0	0	0	0	2	0	2	1	1	0

^aNetworks are named using the target residue.^bNumber of source-target pair networks in which the node is involved.Table 3. Response time of the target residue, the number and distribution of nodes in the different source-target pair signaling networks of the α -subunit.

Source-target pair networks ^a	Residue response time (fs)	Number of nodes	Distribution of the nodes in the source-target pair networks										
			1 ^b	2	3	4	5	6	7	8	9	10	11
Gly150	385	64	15	13	2	5	24	5	0	0	0	0	0
Arg284	35	10	1	2	0	1	0	6	0	0	0	0	0
Ile291	205	51	9	13	0	0	24	5	0	0	0	0	0
Val296	10	2	0	0	0	1	0	1	0	0	0	0	0
Asn299	10	2	1	0	0	0	0	1	0	0	0	0	0
Thr311	10	2	0	1	0	0	0	1	0	0	0	0	0
Lys355	195	39	2	2	1	5	24	5	0	0	0	0	0
Glu366	260	52	11	5	2	5	24	5	0	0	0	0	0
Asn374	195	39	0	4	1	5	24	5	0	0	0	0	0
Ile378	30	6	1	3	0	1	0	1	0	0	0	0	0
Thr380	25	5	0	3	0	1	0	1	0	0	0	0	0

Note: ^aNetworks are named using the target residue.^bNumber of source-target pair networks in which the node is involved.

3.2. Signal transduction networks involved in the intersubunit interactions of the α - and β -subunits upon the binding of threonine molecules

After constructing the protein RRI network, data of residue response time obtained from the energy dissipation simulations were put into the undirected network, resulting in the Initial networks (Figure 3(a) and (b)). The Initial network of the β -subunit consists of 160 nodes and all direction information between them (926 edges). The Initial network of the α -subunit consists of 386 nodes and 2444 edges. With the residues that interact with each other at the binding surface of the two subunits as targets, signal transduction networks for each source-target pair were obtained from the Initial network employing the algorithm of network construction. As shown in Tables 2 and 3, the source-target pair networks consist of different number of nodes. The smallest

network (Val47 and Asn50 as the target residue for the β -subunit; Val296 and Asn299 for the α -subunit) has only 2 nodes, whereas the largest one (Asn125 as the target residue) has 61 nodes for the β -subunit and it is 64 (Gly150 as the target residue) in the case of the α -subunit. Considering the fact that some pathways may be common among them, whereas some may be unique, the numbers of common residues among the source-target pair networks are also listed in Tables 2 and 3. Merging all of the source-target pair networks together results in the 'Union network' (Figure 3(d), (e) and (g)). The Union network is composed of all the possible signaling pathways from the threonine binding site to the intersubunit interaction surface. Change of any residue in the Union network would influence the binding process of the two subunits. However, the underlying mechanism may be different. Some changes may influence the signal

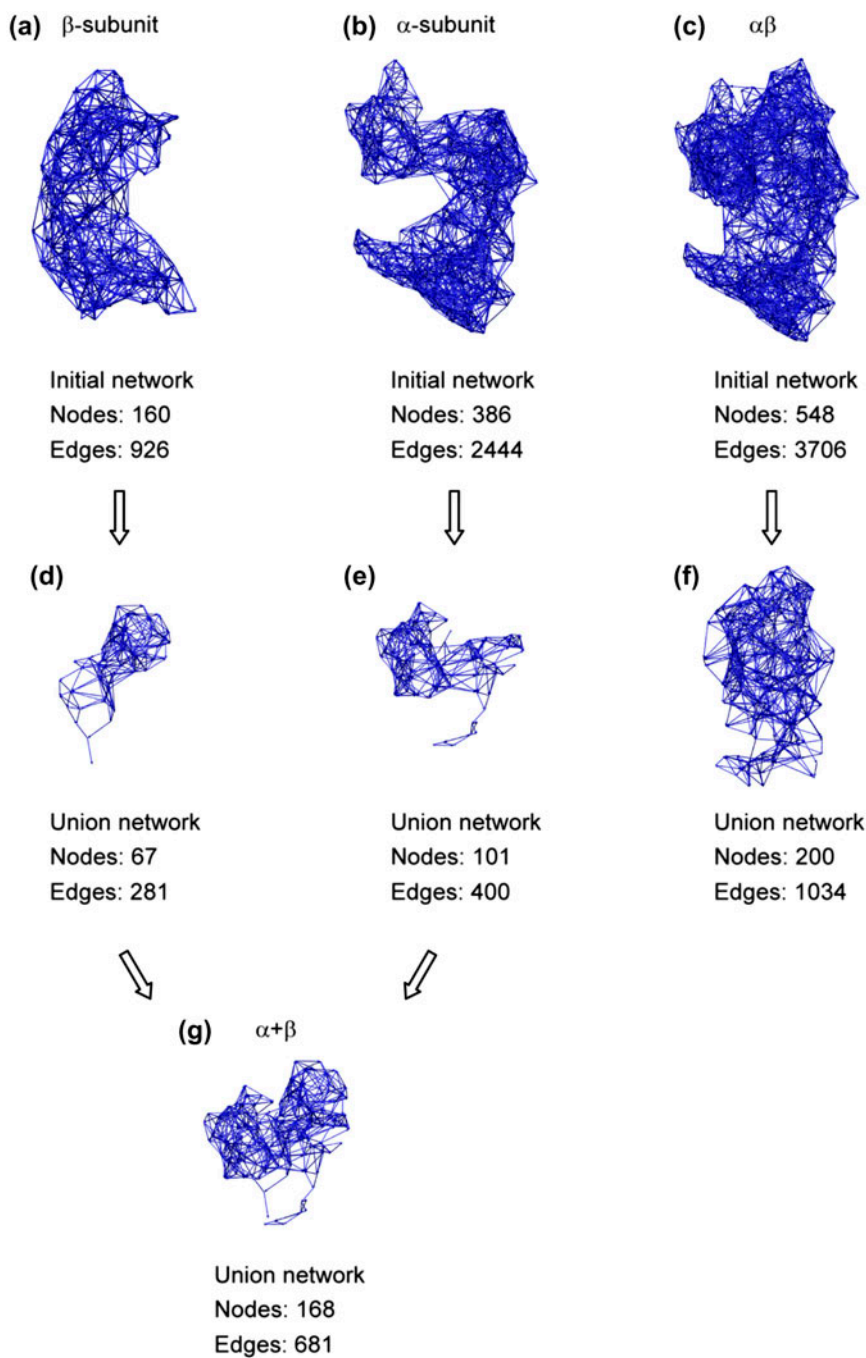


Figure 3. Signal transduction networks. Initial network constructed based on the data when residues residing at the threonine binding pocket of β -subunit (a), α -subunit (b) are perturbed and when residues residing at the lysine binding pocket of CgAK are perturbed (c). Union network constructed based on the data when residues residing at the threonine binding pocket of β -subunit (d), α -subunit (e) are perturbed and when residues residing at the lysine binding pocket of CgAK are perturbed (f). (g) Union network that combine the Union network of both β - and α -subunit.

transduction to only one of the target residues without influencing the others; some changes may affect two or more target residues at the same time (Figure 4(a), (b), (d) and (e)).

In a recent study, an integrated co-evolutionary and molecular dynamics approach, which is based on

sequence information of the protein family and structural information of a number of member proteins, was employed to guide engineering allosteric regulation of *C. glutamicum* aspartokinase (Chen et al., 2011). The allosteric properties were altered as desired to reduce the strength of feedback inhibition by the product(s). How-

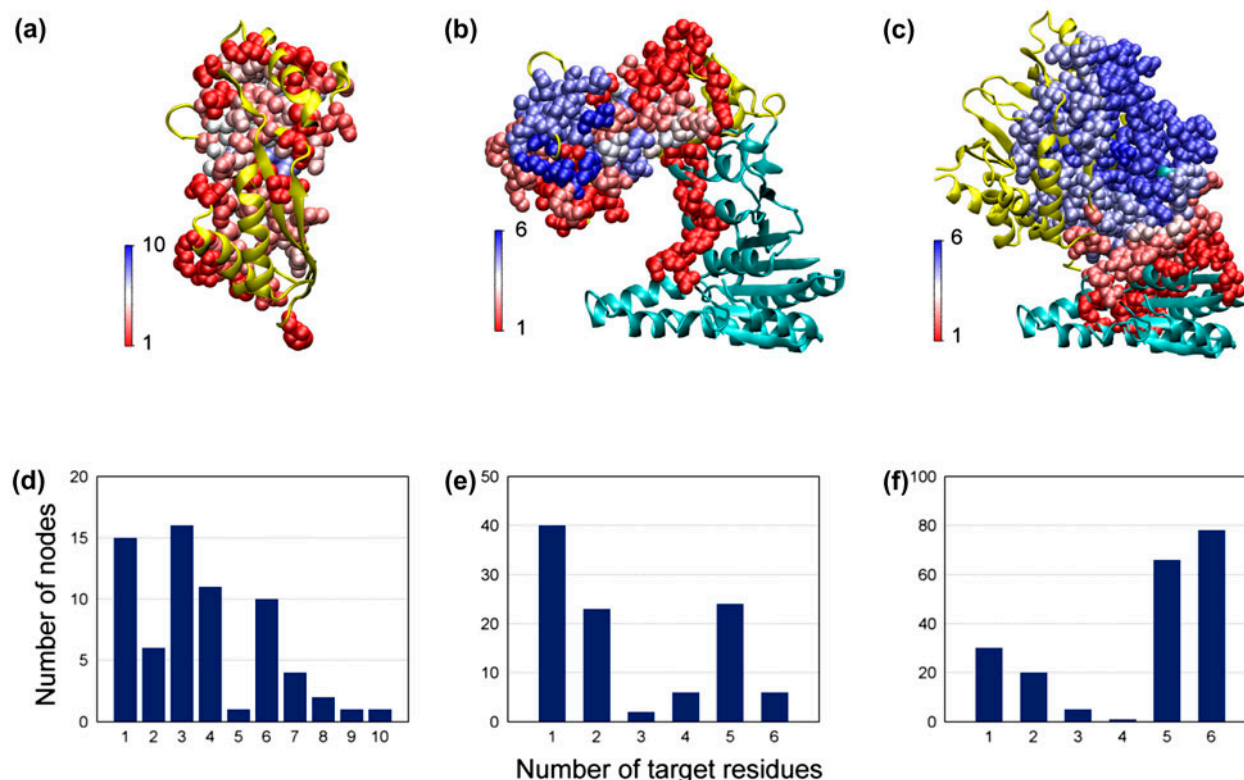


Figure 4. Residues involved in the Union networks. Residues within the Union network of β -subunit (a), α -subunit (b), and CgAK (c) are shaped in VDW surface and colored according to the number of target residues that they can influence. Residues not included in the Union networks are shown in Cartoon; residues belonging to the regulatory domain are colored in yellow; and residues belonging to the catalytic domain are colored in cyan. Distribution of nodes based on the number of target residues that they can influence within the Union network of β -subunit (d), α -subunit (e), and CgAK (f).

ever, the co-evolutionary approach does not provide information for the underlying mechanisms of regulation. Here, the mutation sites predicted by the new approach are compared with those studied by Chen et al. (2011) as well as those that have been experimentally verified by others (Table 4). The Union network, which combines all possible pathways together and thus presents the signal transduction network from the threonine binding site to the intersubunit binding surface, includes all the experimentally proved mutations as well as the mutation sites given by Chen et al. (2011) except F364A. Specially, only one mutation (T361A) is due to the disturbing of the signal transduction network of the β -subunit; two mutations (I291 and N372A) are caused only by the disturbing signaling process within the α -subunit; and the others are because of the disturbing of both the α - and β -subunits (Figure 5(a)). In addition, although most of the mutation sites that influence the β -subunit (63 in 67) also influence the α -subunit, it is not true vice versa (63 in 101). This is because not only residues belonging to the C-terminal regulatory region but also some residues of the N-terminal catalytic domain play roles in the case of the α -subunit.

3.3. Energy dissipation process of CgAK from the lysine binding pocket

In the study of the energy dissipation process within CgAK, energy was input into the key residues of the lysine binding pocket (Asn23(β), Ile24(β), Asp425(β), Thr361(α), and Ser381(α)). As shown in Figure 2(c), the energy dissipation curve exhibits a typical sigmoid pattern that is composed of three stages. The two-state Boltzmann equation was used to describe the energy dissipation (Table 1). For CgAK, half of the residues had been influenced at the time point of 388.7 fs. The half response time is smaller than that when energy was input into the key residues of the threonine binding pocket of the α -subunit, although CgAK is composed of both the α - and β -subunits. This can be explained by the larger dissipation rate constant (1.2 residues fs^{-1} vs. .9 residues fs^{-1}).

In order to get a deeper understanding of the mechanism underlying the process of signal transduction from the lysine binding site of regulatory domain to the catalytic pocket, the response time of each amino acid residue of CgAK was calculated and shown in Figure 2(f) which is colored according to different structural

Table 4. Comparison of mutation sites predicted by the new approach with those reported in literature.

Mutation sites	β	α	$\alpha + \beta$	$\alpha\beta$	Ref	Mutation sites	β	α	$\alpha + \beta$	$\alpha\beta$	References
I272E	✓	✓	✓		Chen et al. (2011)	T311I	✓	✓	✓	✓	Ohnishi et al. (2002)
D274A	✓	✓	✓		Chen et al. (2011)	T336L	✓	✓	✓		Chen et al. 2011
G277A	✓	✓	✓		Yoshida et al. (2007)	T361A	✓		✓	✓	Yoshida et al. (2007)
E278 V	✓	✓	✓		Chen et al. (2011)	F364A				✓	Yoshida et al. (2007)
A279 V	✓	✓	✓		Yoshida et al. (2007)	M365A	✓	✓	✓		Chen et al. (2011)
F283R	✓	✓	✓	✓	Chen et al. (2011)	N372A		✓	✓		Chen et al. (2011)
I291		✓	✓	✓	— ^a	N374A	✓	✓	✓		Chen et al. (2011)
I293	✓	✓	✓	✓	—	I375P	✓	✓	✓	✓	Chen et al. (2011)
D294	✓	✓	✓	✓	—	S381F	✓	✓	✓	✓	Sindelar and Wendisch (2007)
Q298G	✓	✓	✓	✓	Chen et al. (2011)	E382A	✓	✓	✓	✓	Chen et al. (2011)
N299L	✓	✓	✓	✓	Chen et al. (2011)	R384L	✓	✓	✓	✓	Chen et al. (2011)
S301F	✓	✓	✓		Yoshida et al. (2007)	S386A	✓	✓	✓	✓	Chen et al. (2011)
T308A	✓	✓	✓		Yoshida et al. (2007)						

Note: ^aMutations of the corresponding positions have not yet been experimentally proved.

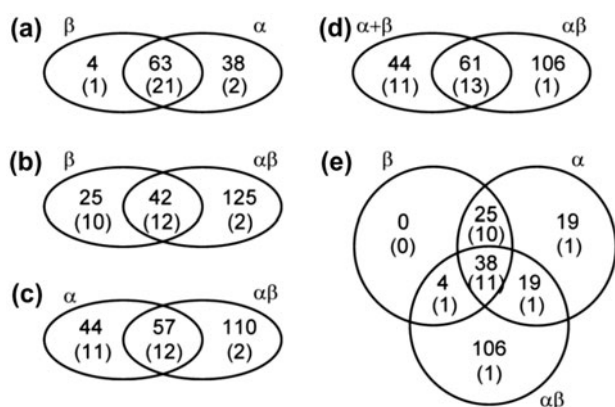


Figure 5. Number of nodes shared by different networks. Number of shared nodes between the Union networks of β -subunit and α -subunit (a), the Union networks of β -subunit and CgAK (b), the Union networks of α -subunit and CgAK (c), the Union networks of β -subunit combined with α -subunit and CgAK (d), and the Union networks of β -subunit, α -subunit, and CgAK (e). Figures in parenthesis are the number of residues that have been experimentally proved.

domains. It is evident that most residues of the regulatory domain (the yellow line) responded to the energy perturbation earlier than those in the catalytic domain (the blue line). It also can be seen that residues belonging to different structural domains may respond to the energy perturbation at the same time point. Residues residing at the catalytic pocket were recognized and their response time was given in Table 5. The response time ranges from 230 fs for Ser209 to 475 for Thr47, which indicate a long signaling process of the allosteric communication considering the fact that all of the residues of the enzyme responded to the energy perturbation within 1000 fs during the energy dissipation simulation.

Table 5. Response time of the target residue, the number and distribution of nodes in the different source-target pair signaling networks of CgAK.

Source-target pair networks ^a	Residues response time (fs)	Number of nodes	Distribution of the nodes in the source-target pair networks					
			1 ^b	2	3	4	5	6
Thr47	475	200	30	20	5	1	66	78
Asp155	395	158	0	10	4	0	66	78
Ser174	375	146	0	0	1	1	66	78
Val176	405	160	0	10	5	1	66	78
Ser209	230	78	0	0	0	0	0	78
Ile211	350	145	0	0	0	1	66	78

Note: ^aNetworks are named using the target residue.

^bNumber of source-target pair networks in which the node is involved.

3.4. Signal transduction network involved in the allosteric communication within CgAK upon the binding of lysine

Data of residue response time obtained from the energy dissipation simulations were put into the protein RRI network of CgAK, resulting in the Initial network (Figure 3(c)). The Initial network consists of 548 nodes and 3706 edges. Considering the fact that the allosteric communication is the process of signal transduction from the regulatory site to the catalytic site, some direction information included in the Initial network may be not relevant to this process and thus could be deleted from the Initial network in order to achieve signal transduction network. For this purpose, residues that interact with substrates were recognized according to the crystal structure of CgAK (PDB code: 3AAW). With these residues as targets, signal transduction networks for each source-target pair were obtained from the Initial network employing the algorithm of network construction. As

shown in Table 5, the smallest network (Ser209 as the target residue) has only 78 nodes, whereas the largest one (Thr47 as the target residue) has 200 nodes. Table 5 also lists the number of common residues among the source–target pair networks. For the largest network, 30 residues are unique; while the others are shared with the other pair networks. The smallest network shares all its residues with the other pair networks. Merging all of the source–target pair networks together results in the ‘Union network’ of CgAK (Figure 3(f)), which is composed of all the possible signaling pathways from the regulatory site to the catalytic site. The number of target residues that can be influenced by the nodes illustrates their different underlying mechanisms (Figure 4(c) and (f)). Moreover, threonines that bind to the two subunits do not exist in the Union network of CgAK.

Mutation sites predicted by the signal transduction network of CgAK are compared with those studied by Chen et al. (2011) and those that have been experimentally verified by others. As shown in Table 6, the Union network of CgAK includes 14 of the 25 mutation sites given by Chen et al. (2011). Specially, two mutations (I291 and F364A) are due to the disturbing of the signal transduction network of the CgAK without disturbing the β -subunit (Figure 5(b)); two mutations (T361A and F364A) are caused by the disturbing of the signal transduction network of the CgAK without disturbing the signaling process within the α -subunit (Figure 5(c)); and only one mutation (F364A) is because of the disturbing of the signal transduction network of the CgAK without disturbing either the β - or α -subunit (Figure 5(d)). It can be seen that an advantage of the new approach is that it is able to provide a deeper insight into the mechanisms underlying the experimental results. In Figure 5(e), the mutation sites predicted by the signal transduction net-

work of CgAK are combined with those given by the networks involved in the intersubunit interactions. Of the 211 sites, 38 sites influence both the binding process of the β - and α -subunits as well as the allosteric process of CgAK; 25 sites influence the binding process of the β - and α -subunits without influencing the allosteric process of CgAK; and 106 sites influence the allosteric process of CgAK without influencing the binding process of the β - and the α -subunits.

To test the statistical significance of this result, the occurrence probability of the result was calculated using the hypergeometric distribution method:

$$p = \frac{\binom{211}{22} \binom{408-211}{211-22}}{\binom{408}{211}} < .01 \quad (4)$$

where p is the occurrence probability; $\binom{n}{m} = \frac{n!}{m!(n-m)!}$ is the binomial coefficient; 408 is the number of residues in the Initial network; 211 is the number of residues in the Union network; and 22 is the number of mutation sites that have been experimentally proved. The small value of the occurrence probability indicates that the result shown here is not a chance and thus those residues predicted by the new method but without experimental prove could also influence the allosteric process.

3.5. Intersubunit signal transduction involved in the allosteric communication within CgAK upon the binding of lysine

CgAK has a unique heteromultimeric structure, which is composed of equimolar α - and β -subunits. The N-terminal

Table 6. Residues involved in the intersubunit signal transduction during the allosteric communication within CgAK.

Residues	Residue response time (fs)	Out-degree	In-degree	IPC	Residues	Residue response time (fs)	Out-degree	In-degree	IPC
Val101 (β)	175	1	1	.02	Val207 (α)	180	1	1	.02
Gly102 (β)	180	2	0	.02	Lys210 (α)	310	0	2	.02
Ala103 (β)	230	3	1	.04	Leu214 (α)	315	0	3	.03
Lys106 (β)	300	1	3	.04	Arg215 (α)	265	0	2	.02
Ile126 (β)	195	0	1	.01	Glu218 (α)	265	0	4	.04
Glu127 (β)	190	0	1	.01	Arg221 (α)	170	1	0	.01
Leu128 (β)	135	1	0	.01	Ser263 (α)	270	1	0	.01
Ser130 (β)	50	3	3	.06	Glu264 (α)	255	1	0	.01
Thr131 (β)	165	2	3	.05	Asp294 (α)	255	1	3	.04
Ser132 (β)	115	4	2	.06	Met295 (α)	235	0	3	.03
Glu133 (β)	205	7	1	.08	Leu297 (α)	125	1	1	.02
Ile134 (β)	165	4	2	.06	Gln298 (α)	180	2	2	.04
Arg135 (β)	45	1	3	.04	Ser379 (α)	75	1	1	.02
Leu202 (α)	195	1	2	.03	Thr380 (α)	20	1	0	.01
Glu203 (α)	105	2	1	.03	Ser381 (α)	30	2	0	.02
Ala205 (α)	270	0	1	.01	Glu382 (α)	15	1	0	.01
Ala206 (α)	235	0	3	.03	Arg384 (α)	35	5	0	.05

portion of α -subunit functions as the catalytic domain, whereas the β -subunit and C-terminal portion of α -subunit serve as the regulatory domain. Thus, there are both intramolecular and intermolecular signal transductions during the process of allosteric regulation of CgAK. Further understanding of the intersubunit signaling process would provide valuable information for the rational design of intermolecular signaling channels.

After identifying the residues that interact with each other at the interface of the two subunits with a contact cutoff of 6 Å, a sub-network composed of nodes that are responsible for the intermolecular communication could be derived from the signal transduction network (Figure 6). Within the 77 residues that transfer signals between the two subunits, less than half of them (34) involve in the process of allosteric communication with 50 edges (Figure 6(a) and (b)). Thereinto, 13 of them belong to the β -subunit, whereas the others belong to the α -subunit (Table 6). With IPC as the parameter to measure the intermolecular propagation capability, it can be found that Glu133 of the β -subunit, which sends signals to seven residues and receive signals from one residue of the α -subunit, shows the largest IPC followed by Ser130, Ser132, and Ile134. For those of the α -subunit, Arg384 has the largest IPC, followed by Glu218, Asp294, and Gln298. The large IPC values of these residues indicate their significance in the process of intersubunit signal transduction. It is noteworthy that the intersubunit signaling is a bi-directional process. This can be seen from the number of in-/out-degree of the nodes. The reason is the regulatory domain of CgAK is constructed by both the β -subunit and the C-terminus of the α -subunit and the lysine binding pocket resides at the cleft between the two structural regions (Yoshida et al., 2010).

The 34 residues that involve in the process of allosteric communication can be further divided into two parts: 21 of them contribute to the signaling process between the β -subunit and the regulatory region of the α -subunit with 26 edges (Figure 6(c)) and 18 of them take part in the signal transduction between the β -subunit and the catalytic domain of the α -subunit with 24 edges (Figure 6(d)). Residues such as Lys106, Ser132, Glu133, Ile134, and Arg135 of the β -subunit contribute both the regulatory region and the catalytic domain of the α -subunit. As for the 21 residues that reside at the α -subunit, 11 belong to the regulatory region of the α -subunit and 10 belong to the catalytic domain of the α -subunit (Tables 7 and 8).

4. Discussion

Perturbation dynamics has been used to discover discrete breathers in protein structures (Juanico, Sanejouand, Piazza, & De Los Rios, 2007; Piazza & Sanejouand, 2008) and long-range energy transfer in proteins (Piazza

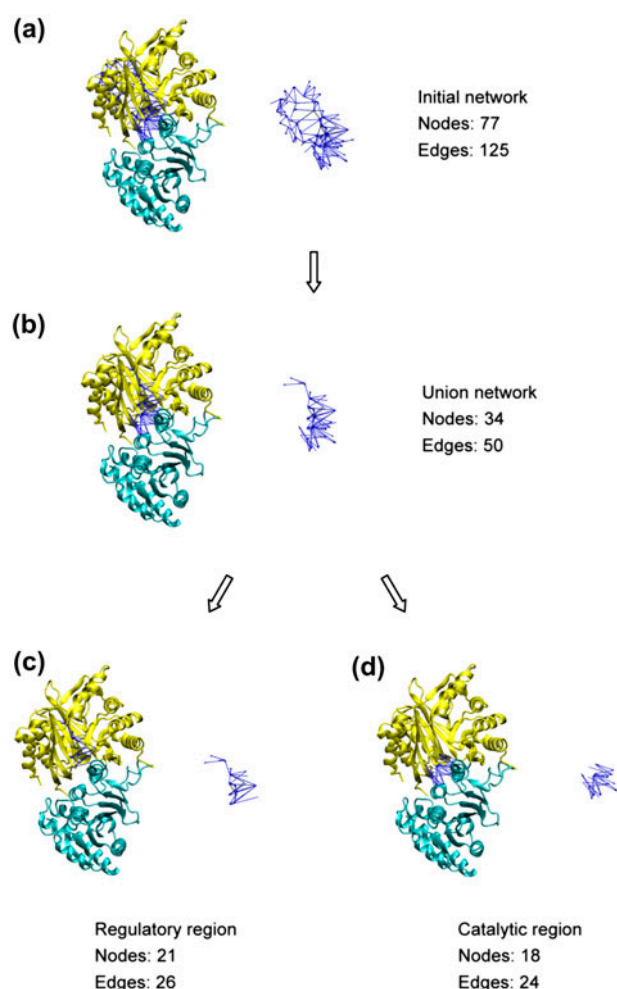


Figure 6. Intersubunit networks. (a) Intersubunit network from the Initial network of CgAK. (b) Intersubunit network from the Union network of CgAK. (c) Intersubunit network between β -subunit and the regulatory region of α -subunit. (d) Intersubunit network between β -subunit and the catalytic region of α -subunit. Residues belonging to the regulatory domain are colored in yellow; residues belonging to the catalytic domain are colored in cyan.

& Sanejouand, 2009). Moreover, perturbation waves can also be applied in proteins and protein networks in signaling and drug design (Antal, Böde, & Csermely, 2009; Farkas et al., 2011). By simulating the input of external energy (binding of an effector) using energy perturbation, protein dynamics is quantitatively described in terms of energy dissipation.

Protein RRI networks have been constructed and studied in literature (Amitai et al., 2004; Vishveshwara, Ghosh, & Hansia, 2009). However, these networks are merely constructed based on the crystal structures of proteins and thus not able to reflect the dynamics characteristics of allosteric proteins. Analysis approaches such as principal component analysis could be combined with

Table 7. Residues involved in the intersubunit signal transduction between β -subunit and the regulatory region of α -subunit.

Residues	Out-degree	In-degree	IPC	Residues	Out-degree	In-degree	IPC
Lys106 (β)	0	3	.06	Ser263 (α)	1	0	.02
Ile126 (β)	0	1	.02	Glu264 (α)	1	0	.02
Glu127 (β)	0	1	.02	Asp294 (α)	1	3	.08
Leu128 (β)	1	0	.02	Met295 (α)	0	3	.06
Ser130 (β)	3	3	.12	Leu297 (α)	1	1	.04
Thr131 (β)	2	3	.10	Gln298 (α)	2	2	.08
Ser132 (β)	2	1	.06	Ser379 (α)	1	1	.04
Glu133 (β)	2	0	.04	Thr380 (α)	1	0	.02
Ile134 (β)	0	1	.02	Ser381 (α)	2	0	.04
Arg135 (β)	0	3	.06	Glu382 (α)	1	0	.02
				Arg384 (α)	5	0	.10

Table 8. Residues involved in the intersubunit signal transduction between β -subunit and the catalytic region of α -subunit.

Residues	Out-degree	In-degree	IPC	Residues	Out-degree	In-degree	IPC
Val101 (β)	1	1	.04	Leu202 (α)	1	2	.06
Gly102 (β)	2	0	.04	Glu203 (α)	2	1	.06
Ala103 (β)	3	1	.08	Ala205 (α)	0	1	.02
Lys106 (β)	1	0	.02	Ala206 (α)	0	3	.06
Ser132 (β)	2	1	.06	Val207 (α)	1	1	.04
Glu133 (β)	5	1	.12	Lys210 (α)	0	2	.04
Ile134 (β)	4	1	.10	Leu214 (α)	0	3	.06
Arg135 (β)	1	0	.02	Arg215 (α)	0	2	.04
				Glu218 (α)	0	4	.08
				Arg221 (α)	1	0	.02

traditional molecular dynamics simulations to find signaling-related residues. Recently, protein structure network and community network analysis have been developed to investigate the protein conformational changes and identify the signaling pathways in multimeric proteins (Hansia, Ghosh, & Vishveshwara, 2009; Sethi, Eargle, Black, & Luthey-Schulten, 2009). Nevertheless, the information flow within the protein complex upon binding of effectors is still unclear.

In order to explore how the signal is transferred within molecules like allosteric proteins upon binding of an effector, a novel algorithm of network construction based on the energy dissipation model is proposed in this work to reveal the dynamic process of intra- and inter-molecular signal transduction related to allosteric regulation. A key feature of this approach is that direction information is specified after inferring RRI network involved in the process of signal transduction. This enables fundamental analysis of the regulation hierarchy and identification of regulation hubs of the signaling network with the help of network analysis tools (Szalay-Beko et al., 2012). A further advantage of the new approach presented in this work is that it gives a deeper insight into the mechanisms of allosteric proteins, thus providing useful information that is of particular interest for redesign of allosteric regulation in

bioengineering and even development of effectors for disease-related allosteric proteins in biomedicine.

From the model system CgAK studied in this work, it can be seen that the energy dissipation curve of the residue response time for the C-terminus of the α -subunit shows a similar pattern to that for the β -subunit, although the α -subunit is composed of both the C-terminal regulatory region, which shares the same gene sequence with the β -subunit and the N-terminal catalytic domain. The similarity is due to the fact that these two structural regions bind to each other in a symmetric manner to compose the regulatory domain and threonine molecules bind to each structural region in the same way (Yoshida et al., 2010). This dynamic character discovered by the energy dissipation model indicates the relative independence between the C-terminal regulatory region and the N-terminal catalytic domain of the α -subunit.

It is known that CgAK is regulated by threonine and lysine in a concerted manner. However, from the signal transduction network constructed for CgAK, it is found that threonine molecules that bind to the two subunits do not exist in the Union network of CgAK, indicating that the function of these threonine molecules is to facilitate the binding of the two subunits without contributing to the allosteric communication. Furthermore, from the results of

intersubunit signal transduction involved in the allosteric communication within CgAK, it can be seen that during the allosteric communication, signals can be transferred from the β -subunit to the catalytic region of the α -subunit through both a direct way of intersubunit signal transduction and an indirect way: first, to the regulatory region of the α -subunit by intersubunit signal transduction and then to the catalytic region by intramolecular signal transduction.

Funding

This work was supported by grants from the China Scholarship Council (to CWM). CWM and APZ were also supported by the German Research Foundation (DFG) through the project ZE 542/6-1 and the Hamburg Excellence Initiative project SynBio.

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